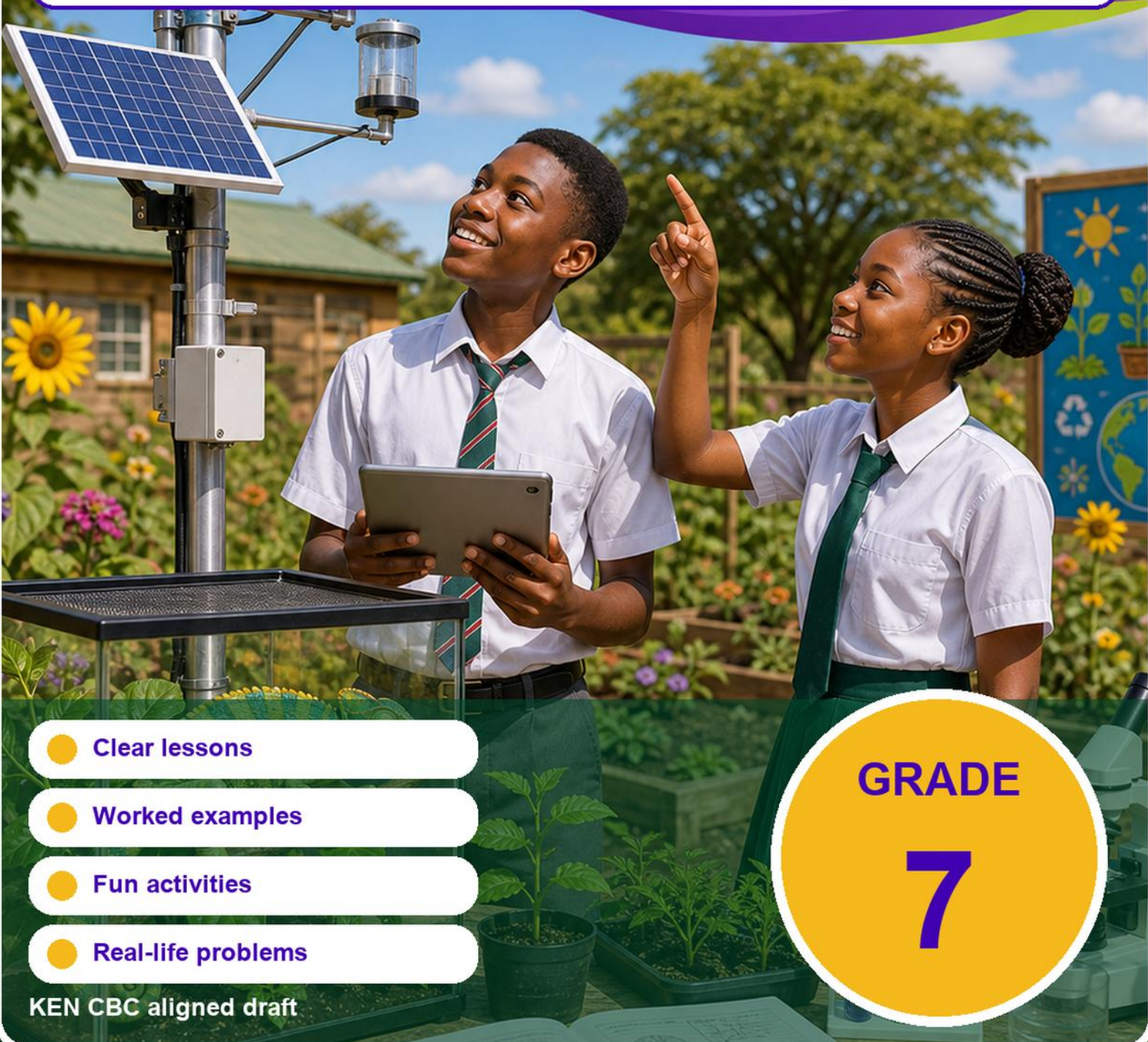


KITABU QUEST

Grade 7

SCIENCE AND TECHNOLOGY



● Clear lessons

● Worked examples

● Fun activities

● Real-life problems

KEN CBC aligned draft

GRADE

7

Title Page

KITABU QUEST Grade 7 Science and Technology

Kenya CBC content draft for review

Mascot: chameleon

This content-first draft was regenerated from Kitabu curriculum data using a topic-by-topic book plan. It is designed to help learners read, practise, discuss, create, and revise with confidence.

Cover artwork is intentionally deferred until all content has passed review and is marked ready-for-cover.

How To Use This Book

This book helps Grade 7 learners study Science and Technology through clear lessons, worked examples, activities, practice, and revision.

Use the inquiry questions to think first. Read the explanation. Try the guided example. Complete the activity. Answer the practice questions. Correct your work and ask for support where needed.

Observe, ask questions, test safely, record evidence, and explain findings.

Every unit has a local example, a misconception to avoid, success criteria, and a check for understanding.

Learning Skills In This Book

Each chapter builds knowledge, skills, values, creativity, communication, collaboration, critical thinking, digital literacy, and self-confidence.

Look for these page types: Learn, Example, Activity, Practice, Reflection, Home Link, and Review.

How to study well: read the goal first, underline new words, try the example before reading the answer, discuss with a partner, and correct your work after feedback. Do not skip the reflection questions. They help you notice what you understand and what needs more practice.

Table Of Contents

Use this table of contents to plan your study. Start with the first chapter, but return to earlier pages whenever you need revision.

1. 1.0 Scientific Investigation
2. 2.0 Mixtures, Elements, and Compounds
3. 3.0 Living Things and their Environment
4. 4.0 Force and Energy

After each chapter, complete the chapter review before moving forward. At the end of the book, use the glossary, answer notes, and final project to revise the whole subject.

Chapter: Scientific Investigation

In this chapter you will study Scientific Investigation.

Inquiry starter: What do you already know about this topic from home, school, or your community?

Chapter units:

- 1.1 Introduction to Integrated Science
- 1.2 Laboratory Safety
- 1.3 Laboratory apparatus and instruments

Before you begin, write two things you already know and one question you want to answer. As you study, collect examples from daily life. At the end of the chapter, return to your question and improve your answer using new vocabulary, examples, and evidence from the lessons.

Introduction to Integrated Science: Lesson Opener

Learning area: Introduction to Integrated Science

Start here: a safe observation at home, school, garden, workshop, health club, or local environment linked to Introduction to Integrated Science.

Curriculum link: This lesson follows the Grade 7 curriculum sequence and focuses on Introduction to Integrated Science.

By the end of this lesson sequence, you should be able to:

- outline the components of Integrated Science as a field of study
- explain the importance of science in daily life
- show interest in learning Integrated Science at junior school.

Inquiry questions:

- How is the knowledge acquired in Integrated Science useful in daily life?

Before reading, write one thing you already know and one question you want this lesson to answer.

Introduction to Integrated Science: Learn

Start with this real situation: a safe observation at home, school, garden, workshop, health club, or local environment linked to Introduction to Integrated Science.

Important idea: Introduction to Integrated Science.

Science and technology learning starts with a question. A useful answer is built from evidence: observations, measurements, diagrams, models, tests, data tables, or design trials. Separate what you saw from what you think it means.

A common mistake is writing guesses as facts without observing, testing safely, or recording evidence. Avoid it by separating what you saw from what you think it means.

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

For Grade 7, use labelled sketches, tables, and short evidence-based explanations.

Inline visual support:

| Evidence tool | How to use it in Introduction to Integrated Science |

| --- | --- |

| Labelled sketch | Name visible parts and show direction or change with arrows. |

| Observation table | Record facts before explaining them. |

| Safety note | Name the hazard and the safe action. |

| Conclusion | Begin with: The evidence shows. |

Method for Science and Technology: Observe, ask questions, test safely, record evidence, and explain findings.

Key words:

- Introduction to Integrated Science
- Introduction
- Integrated
- Science

Explain the idea in your own words before attempting the activities.

Introduction to Integrated Science: Worked Example

Investigation model for Introduction to Integrated Science: Move from question to evidence.

Question: What evidence would help explain this idea clearly?

Procedure:

1. State the exact question in one sentence.
2. Name the materials, diagram, data, model, or digital tool needed.
3. Decide what will be observed, measured, compared, or designed.
4. Record results in a table, labelled sketch, flow chart, or screenshot note.
5. Explain what the evidence means and name one limit of the investigation.

Evidence table:

| Step | Evidence collected | What it shows |

| --- | --- | --- |

| Observe or test | specific detail | explanation |

Conclusion frame: The evidence supports this answer because _____.

Introduction to Integrated Science: Activity

Activity for Introduction to Integrated Science: Plan a safe observation or investigation.

Inquiry question: How is the knowledge acquired in Integrated Science useful in daily life?

1. State the question you want to answer.
2. List safe materials and one safety rule.
3. Draw a labelled setup or observation table before starting.
4. Record at least three observations. Do not write guesses as facts.
5. Compare evidence with a partner and discuss what the evidence means.
6. Write a short conclusion that begins, "The evidence shows."

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

Home link: Look for a safe related example at home or in the community. Record what you observed without disturbing people, animals, plants, or devices.

Introduction to Integrated Science: Activity And Practice

Practice for Introduction to Integrated Science:

Use this routine for every answer: question, materials or evidence, method, observation, conclusion, and safety.

1. Complete a table with three components of Integrated Science: Biology, Chemistry, and Physics. For each component, write one school or home example: living things, materials, and energy or forces. Answer guide: Biology studies living things such as plants; Chemistry studies substances such as water and salts; Physics studies energy, forces, light, heat, and electricity. Target outcome: outline the components of Integrated Science as a field of study. Include labels, evidence or expected observation, a safety note where relevant, and a conclusion.
2. Choose three daily-life examples where science helps people: clean water, safe cooking fuel, crop growth, medicine, transport, or electricity. For each example, name the science idea and the benefit. Answer guide: clean water uses filtration or boiling to reduce disease; farming uses soil and plant knowledge; electricity uses circuits and energy transfer. Target outcome: explain the importance of science in daily life. Include labels, evidence or expected observation, a safety note where relevant, and a conclusion.
3. Create a junior-school science learning plan with one question from Biology, one from Chemistry, and one from Physics. State how each question can be investigated safely. Answer guide: useful questions are observable or testable, use safe materials, and lead to recorded evidence. Target outcome: show interest in learning Integrated Science at junior school. Include labels, evidence or expected observation, a safety note where relevant, and a conclusion.

Correction check:

- observation is specific
- safety is considered
- conclusion uses evidence

Reflection: Which part of your answer is evidence? Which part is explanation?

Home link: Observe a safe related example at home or in the community. Record facts only, without disturbing people, animals, plants, or devices.

Introduction to Integrated Science: Outcome Check

Outcome: outline the components of Integrated Science as a field of study

Show mastery with topic evidence.

Task: Complete a table with three components of Integrated Science: Biology, Chemistry, and Physics. For each component, write one school or home example: living things, materials, and energy or forces. Answer guide: Biology studies living things such as plants; Chemistry studies substances such as water and salts; Physics studies energy, forces, light, heat, and electricity.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows." or "The model shows."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Introduction to Integrated Science, uses correct science vocabulary, and connects evidence to explanation.

Introduction to Integrated Science: Outcome Check

Outcome: explain the importance of science in daily life

Show mastery with topic evidence.

Task: Choose three daily-life examples where science helps people: clean water, safe cooking fuel, crop growth, medicine, transport, or electricity. For each example, name the science idea and the benefit. Answer guide: clean water uses filtration or boiling to reduce disease; farming uses soil and plant knowledge; electricity uses circuits and energy transfer.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows." or "The model shows."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Introduction to Integrated Science, uses correct science vocabulary, and connects evidence to explanation.

Introduction to Integrated Science: Outcome Check

Outcome: show interest in learning Integrated Science at junior school.

Show mastery with topic evidence.

Task: Create a junior-school science learning plan with one question from Biology, one from Chemistry, and one from Physics. State how each question can be investigated safely. Answer guide: useful questions are observable or testable, use safe materials, and lead to recorded evidence.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows." or "The model shows."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Introduction to Integrated Science, uses correct science vocabulary, and connects evidence to explanation.

Laboratory Safety: common hazards and their symbols in the laboratory: Lesson Opener

Learning area: Laboratory Safety: common hazards and their symbols in the laboratory

Start here: a safe observation at home, school, garden, workshop, health club, or local environment linked to Laboratory Safety: common hazards and their symbols in the laboratory.

Curriculum link: This lesson follows the Grade 7 curriculum sequence and focuses on Laboratory Safety: common hazards and their symbols in the laboratory.

By the end of this lesson sequence, you should be able to:

- identify common hazards and their symbols in the laboratory
- explain the causes of common accidents in the laboratory
- describe First Aid measures for common laboratory accidents

Inquiry questions:

- What safety measures should be considered while working in the laboratory?

Before reading, write one thing you already know and one question you want this lesson to answer.

Laboratory Safety: common hazards and their symbols in the laboratory: Learn

Start with this real situation: a safe observation at home, school, garden, workshop, health club, or local environment linked to Laboratory Safety: common hazards and their symbols in the laboratory.

Important idea: Laboratory Safety: common hazards and their symbols in the laboratory.

Science and technology learning starts with a question. A useful answer is built from evidence: observations, measurements, diagrams, models, tests, data tables, or design trials. Separate what you saw from what you think it means.

A common mistake is writing guesses as facts without observing, testing safely, or recording evidence. Avoid it by separating what you saw from what you think it means.

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

For Grade 7, use labelled sketches, tables, and short evidence-based explanations.

Inline visual support:

| Evidence tool | How to use it in Laboratory Safety: common hazards and their symbols in the laboratory |

| --- | --- |

| Labelled sketch | Name visible parts and show direction or change with arrows. |

| Observation table | Record facts before explaining them. |

| Safety note | Name the hazard and the safe action. |

| Conclusion | Begin with: The evidence shows. |

Method for Science and Technology: Observe, ask questions, test safely, record evidence, and explain findings.

Key words:

- Laboratory Safety: common hazards and their symbols in the laboratory
- Laboratory Safety
- Laboratory
- Safety
- common
- hazards
- their
- symbols

Explain the idea in your own words before attempting the activities.

Laboratory Safety: common hazards and their symbols in the laboratory: Worked Example

Investigation model for Laboratory Safety: common hazards and their symbols in the laboratory: Move from question to evidence.

Question: What evidence would help explain this idea clearly?

Procedure:

1. State the exact question in one sentence.
2. Name the materials, diagram, data, model, or digital tool needed.
3. Decide what will be observed, measured, compared, or designed.
4. Record results in a table, labelled sketch, flow chart, or screenshot note.
5. Explain what the evidence means and name one limit of the investigation.

Evidence table:

Step	Evidence collected	What it shows
---	---	---
Observe or test	specific detail	explanation

Conclusion frame: The evidence supports this answer because _____.

Laboratory Safety: common hazards and their symbols in the laboratory: Activity

Activity for Laboratory Safety: common hazards and their symbols in the laboratory: Plan a safe observation or investigation.

Inquiry question: What safety measures should be considered while working in the laboratory?

1. State the question you want to answer.
2. List safe materials and one safety rule.
3. Draw a labelled setup or observation table before starting.
4. Record at least three observations. Do not write guesses as facts.
5. Compare evidence with a partner and discuss what the evidence means.
6. Write a short conclusion that begins, "The evidence shows."

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

Home link: Look for a safe related example at home or in the community. Record what you observed without disturbing people, animals, plants, or devices.

Laboratory Safety: common hazards and their symbols in the laboratory: Activity And Practice

Practice for Laboratory Safety: common hazards and their symbols in the laboratory:

Use this routine for every answer: question, materials or evidence, method, observation, conclusion, and safety.

1. Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome. Target outcome: identify common hazards and their symbols in the laboratory. Include labels, evidence or expected observation, a safety note where relevant, and a conclusion.
2. Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome. Target outcome: explain the causes of common accidents in the laboratory. Include labels, evidence or expected observation, a safety note where relevant, and a conclusion.
3. Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome. Target outcome: describe First Aid measures for common laboratory accidents. Include labels, evidence or expected observation, a safety note where relevant, and a conclusion.

Correction check:

- observation is specific
- safety is considered
- conclusion uses evidence

Reflection: Which part of your answer is evidence? Which part is explanation?

Home link: Observe a safe related example at home or in the community. Record facts only, without disturbing people, animals, plants, or devices.

Laboratory Safety: common hazards and their symbols in the laboratory: Outcome Check

Outcome: identify common hazards and their symbols in the laboratory

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows." or "The model shows."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Laboratory Safety: common hazards and their symbols in the laboratory, uses correct science vocabulary, and connects evidence to explanation.

Laboratory Safety: common hazards and their symbols in the laboratory: Outcome Check

Outcome: explain the causes of common accidents in the laboratory

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows." or "The model shows."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Laboratory Safety: common hazards and their symbols in the laboratory, uses correct science vocabulary, and connects evidence to explanation.

Laboratory Safety: common hazards and their symbols in the laboratory: Outcome Check

Outcome: describe First Aid measures for common laboratory accidents

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows." or "The model shows."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Laboratory Safety: common hazards and their symbols in the laboratory, uses correct science vocabulary, and connects evidence to explanation.

Laboratory Safety: in the laboratory and access: Lesson Opener

Learning area: Laboratory Safety: in the laboratory and access

Start here: a safe observation at home, school, garden, workshop, health club, or local environment linked to Laboratory Safety: in the laboratory and access.

Curriculum link: This lesson follows the Grade 7 curriculum sequence and focuses on Laboratory Safety: in the laboratory and access.

By the end of this lesson sequence, you should be able to:

- safety in the laboratory and access to a healthy working environment

Inquiry questions:

- What safety measures should be considered while working in the laboratory?

Before reading, write one thing you already know and one question you want this lesson to answer.

Laboratory Safety: in the laboratory and access: Learn

Start with this real situation: a safe observation at home, school, garden, workshop, health club, or local environment linked to Laboratory Safety: in the laboratory and access.

Important idea: Laboratory Safety: in the laboratory and access.

Environmental science connects observation with care for people and living things. Record place, date, conditions, and evidence. Then explain the pattern and propose a practical improvement that protects health, safety, food, water, or the environment.

A common mistake is writing guesses as facts without observing, testing safely, or recording evidence. Avoid it by separating what you saw from what you think it means.

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

For Grade 7, use labelled sketches, tables, and short evidence-based explanations.

Inline visual support:

Evidence tool	How to use it in Laboratory Safety: in the laboratory and access
Labelled sketch	Name visible parts and show direction or change with arrows.
Observation table	Record facts before explaining them.
Safety note	Name the hazard and the safe action.
Conclusion	Begin with: The evidence shows.

Method for Science and Technology: Observe, ask questions, test safely, record evidence, and explain findings.

Key words:

- Laboratory Safety: in the laboratory and access
- Laboratory Safety
- Laboratory
- Safety
- access

Explain the idea in your own words before attempting the activities.

Laboratory Safety: in the laboratory and access: Worked Example

Investigation model for Laboratory Safety: in the laboratory and access: Use field observation carefully.

Question: What evidence in the local environment shows the pattern or problem?

Example: A school garden with dry soil, wilting leaves, and cracks near the roots shows that plants need steady water and soil cover to reduce moisture loss.

Procedure:

1. Choose a safe observation point in the school compound, garden, home area, or community.
2. Record date, weather, place, and what you can see without disturbing living things.
3. Draw a labelled sketch or complete a three-column observation table.
4. Separate facts from explanations.
5. Suggest one practical improvement that protects health, safety, food, water, or the environment.

Conclusion frame: My observations show _____ because the evidence was _____.

Laboratory Safety: in the laboratory and access: Activity

Activity for Laboratory Safety: in the laboratory and access: Plan a safe observation or investigation.

Inquiry question: What safety measures should be considered while working in the laboratory?

1. State the question you want to answer.
2. List safe materials and one safety rule.
3. Draw a labelled setup or observation table before starting.
4. Record at least three observations. Do not write guesses as facts.
5. Compare evidence with a partner and discuss what the evidence means.
6. Write a short conclusion that begins, "The evidence shows."

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

Home link: Look for a safe related example at home or in the community. Record what you observed without disturbing people, animals, plants, or devices.

Laboratory Safety: in the laboratory and access: Activity And Practice

Practice for Laboratory Safety: in the laboratory and access:

Use this routine for every answer: question, materials or evidence, method, observation, conclusion, and safety.

1. Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome. Target outcome: safety in the laboratory and access to a healthy working environment. Include labels, evidence or expected observation, a safety note where relevant, and a conclusion.

Correction check:

- observation is specific
- safety is considered
- conclusion uses evidence

Reflection: Which part of your answer is evidence? Which part is explanation?

Home link: Observe a safe related example at home or in the community. Record facts only, without disturbing people, animals, plants, or devices.

Laboratory Safety: in the laboratory and access: Outcome Check

Outcome: safety in the laboratory and access to a healthy working environment

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows." or "The model shows."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Laboratory Safety: in the laboratory and access, uses correct science vocabulary, and connects evidence to explanation.

Laboratory apparatus and instruments: basic practise use and care for apparatus skills: Lesson Opener

Learning area: Laboratory apparatus and instruments: basic practise use and care for apparatus skills

Start here: a safe observation at home, school, garden, workshop, health club, or local environment linked to Laboratory apparatus and instruments: basic practise use and care for apparatus skills.

Curriculum link: This lesson follows the Grade 7 curriculum sequence and focuses on Laboratory apparatus and instruments: basic practise use and care for apparatus skills.

By the end of this lesson sequence, you should be able to:

- describe the basic practise use and care for apparatus skills in science
- use and care for apparatus and instruments in the laboratory
- use the SI units for basic and derived quantities collaboratively carry out activities in science

Inquiry questions:

- Why are basic science skills important?
- How are quantities in science expressed?

Before reading, write one thing you already know and one question you want this lesson to answer.

Laboratory apparatus and instruments: basic practise use and care for apparatus skills: Learn

Start with this real situation: a safe observation at home, school, garden, workshop, health club, or local environment linked to Laboratory apparatus and instruments: basic practise use and care for apparatus skills.

Important idea: Laboratory apparatus and instruments: basic practise use and care for apparatus skills.

Science and technology learning starts with a question. A useful answer is built from evidence: observations, measurements, diagrams, models, tests, data tables, or design trials. Separate what you saw from what you think it means.

A common mistake is writing guesses as facts without observing, testing safely, or recording evidence. Avoid it by separating what you saw from what you think it means.

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

For Grade 7, use labelled sketches, tables, and short evidence-based explanations.

Inline visual support:

| Evidence tool | How to use it in Laboratory apparatus and instruments: basic practise use and care for apparatus skills |

| --- | --- |

| Labelled sketch | Name visible parts and show direction or change with arrows. |

| Observation table | Record facts before explaining them. |

| Safety note | Name the hazard and the safe action. |

| Conclusion | Begin with: The evidence shows. |

Method for Science and Technology: Observe, ask questions, test safely, record evidence, and explain findings.

Key words:

- Laboratory apparatus and instruments: basic practise use and care for apparatus skills
- Laboratory apparatus and instruments
- Laboratory
- apparatus
- instruments
- basic
- practise
- skills

Explain the idea in your own words before attempting the activities.

Laboratory apparatus and instruments: basic practise use and care for apparatus skills: Worked Example

Investigation model for Laboratory apparatus and instruments: basic practise use and care for apparatus skills: Move from question to evidence.

Question: What evidence would help explain this idea clearly?

Procedure:

1. State the exact question in one sentence.
2. Name the materials, diagram, data, model, or digital tool needed.
3. Decide what will be observed, measured, compared, or designed.
4. Record results in a table, labelled sketch, flow chart, or screenshot note.
5. Explain what the evidence means and name one limit of the investigation.

Evidence table:

Step	Evidence collected	What it shows
---	---	---
Observe or test	specific detail	explanation

Conclusion frame: The evidence supports this answer because _____.

Laboratory apparatus and instruments: basic practise use and care for apparatus skills: Activity

Activity for Laboratory apparatus and instruments: basic practise use and care for apparatus skills: Plan a safe observation or investigation.

Inquiry question: Why are basic science skills important?

1. State the question you want to answer.
2. List safe materials and one safety rule.
3. Draw a labelled setup or observation table before starting.
4. Record at least three observations. Do not write guesses as facts.
5. Compare evidence with a partner and discuss what the evidence means.
6. Write a short conclusion that begins, "The evidence shows."

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

Home link: Look for a safe related example at home or in the community. Record what you observed without disturbing people, animals, plants, or devices.

Laboratory apparatus and instruments: basic practise use and care for apparatus skills: Activity And Practice

Practice for Laboratory apparatus and instruments: basic practise use and care for apparatus skills:

Use this routine for every answer: question, materials or evidence, method, observation, conclusion, and safety.

1. Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome. Target outcome: describe the basic practise use and care for apparatus skills in science. Include labels, evidence or expected observation, a safety note where relevant, and a conclusion.
2. Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome. Target outcome: use and care for apparatus and instruments in the laboratory. Include labels, evidence or expected observation, a safety note where relevant, and a conclusion.
3. Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome. Target outcome: use the SI units for basic and derived quantities collaboratively carry out activities in science. Include labels, evidence or expected observation, a safety note where relevant, and a conclusion.

Correction check:

- observation is specific
- safety is considered
- conclusion uses evidence

Reflection: Which part of your answer is evidence? Which part is explanation?

Home link: Observe a safe related example at home or in the community. Record facts only, without disturbing people, animals, plants, or devices.

Laboratory apparatus and instruments: basic practise use and care for apparatus skills: Outcome Check

Outcome: describe the basic practise use and care for apparatus skills in science

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows." or "The model shows."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Laboratory apparatus and instruments: basic practise use and care for apparatus skills, uses correct science vocabulary, and connects evidence to explanation.

Laboratory apparatus and instruments: basic practise use and care for apparatus skills: Outcome Check

Outcome: use and care for apparatus and instruments in the laboratory

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows." or "The model shows."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Laboratory apparatus and instruments: basic practise use and care for apparatus skills, uses correct science vocabulary, and connects evidence to explanation.

Laboratory apparatus and instruments: basic practise use and care for apparatus skills: Outcome Check

Outcome: use the SI units for basic and derived quantities collaboratively carry out activities in science

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows." or "The model shows."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Laboratory apparatus and instruments: basic practise use and care for apparatus skills, uses correct science vocabulary, and connects evidence to explanation.

Laboratory apparatus and instruments: consumer protection when handling different apparatus, instruments: Lesson Opener

Learning area: Laboratory apparatus and instruments: consumer protection when handling different apparatus, instruments

Start here: a safe observation at home, school, garden, workshop, health club, or local environment linked to Laboratory apparatus and instruments: consumer protection when handling different apparatus, instruments,.

Curriculum link: This lesson follows the Grade 7 curriculum sequence and focuses on Laboratory apparatus and instruments: consumer protection when handling different apparatus, instruments.

By the end of this lesson sequence, you should be able to:

- appreciate consumer protection when handling different apparatus, instruments, and work with peers to carry out other

Inquiry questions:

- How are quantities in science expressed?

Before reading, write one thing you already know and one question you want this lesson to answer.

Laboratory apparatus and instruments: consumer protection when handling different apparatus, instruments: Learn

Start with this real situation: a safe observation at home, school, garden, workshop, health club, or local environment linked to Laboratory apparatus and instruments: consumer protection when handling different apparatus, instruments,.

Important idea: Laboratory apparatus and instruments: consumer protection when handling different apparatus, instruments,.

Science and technology learning starts with a question. A useful answer is built from evidence: observations, measurements, diagrams, models, tests, data tables, or design trials. Separate what you saw from what you think it means.

A common mistake is writing guesses as facts without observing, testing safely, or recording evidence. Avoid it by separating what you saw from what you think it means.

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

For Grade 7, use labelled sketches, tables, and short evidence-based explanations.

Inline visual support:

| Evidence tool | How to use it in Laboratory apparatus and instruments: consumer protection when handling different apparatus, instruments, |

| --- | --- |

| Labelled sketch | Name visible parts and show direction or change with arrows. |

| Observation table | Record facts before explaining them. |

| Safety note | Name the hazard and the safe action. |

| Conclusion | Begin with: The evidence shows. |

Method for Science and Technology: Observe, ask questions, test safely, record evidence, and explain findings.

Key words:

- Laboratory apparatus and instruments: consumer protection when handling different apparatus, instruments
- Laboratory apparatus and instruments
- Laboratory
- apparatus
- instruments
- consumer
- protection
- handling

Explain the idea in your own words before attempting the activities.

Laboratory apparatus and instruments: consumer protection when handling different apparatus, instruments: Worked Example

Investigation model for Laboratory apparatus and instruments: consumer protection when handling different apparatus, instruments,: Move from question to evidence.

Question: What evidence would help explain this idea clearly?

Procedure:

1. State the exact question in one sentence.
2. Name the materials, diagram, data, model, or digital tool needed.
3. Decide what will be observed, measured, compared, or designed.
4. Record results in a table, labelled sketch, flow chart, or screenshot note.
5. Explain what the evidence means and name one limit of the investigation.

Evidence table:

Step	Evidence collected	What it shows
---	---	---
Observe or test	specific detail	explanation

Conclusion frame: The evidence supports this answer because _____.

Laboratory apparatus and instruments: consumer protection when handling different apparatus, instruments: Activity

Activity for Laboratory apparatus and instruments: consumer protection when handling different apparatus, instruments,: Plan a safe observation or investigation.

Inquiry question: How are quantities in science expressed?

1. State the question you want to answer.
2. List safe materials and one safety rule.
3. Draw a labelled setup or observation table before starting.
4. Record at least three observations. Do not write guesses as facts.
5. Compare evidence with a partner and discuss what the evidence means.
6. Write a short conclusion that begins, "The evidence shows."

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

Home link: Look for a safe related example at home or in the community. Record what you observed without disturbing people, animals, plants, or devices.

Laboratory apparatus and instruments: consumer protection when handling different apparatus, instruments: Activity And Practice

Practice for Laboratory apparatus and instruments: consumer protection when handling different apparatus, instruments:

Use this routine for every answer: question, materials or evidence, method, observation, conclusion, and safety.

1. Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome. Target outcome: appreciate consumer protection when handling different apparatus, instruments, and work with peers to carry out other. Include labels, evidence or expected observation, a safety note where relevant, and a conclusion.

Correction check:

- observation is specific
- safety is considered
- conclusion uses evidence

Reflection: Which part of your answer is evidence? Which part is explanation?

Home link: Observe a safe related example at home or in the community. Record facts only, without disturbing people, animals, plants, or devices.

Laboratory apparatus and instruments: consumer protection when handling different apparatus, instruments: Outcome Check

Outcome: appreciate consumer protection when handling different apparatus, instruments, and work with peers to carry out other

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows." or "The model shows."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Laboratory apparatus and instruments: consumer protection when handling different apparatus, instruments,, uses correct science vocabulary, and connects evidence to explanation.

Scientific Investigation: Chapter Review

Review the chapter carefully.

1. Write five key words from this chapter and explain each one.
2. Choose two units and compare what you learned.
3. Create one poster, chart, dialogue, model, map, or worked solution that teaches this chapter.
4. Answer one inquiry question again. Improve your first answer using what you now know.
5. Rate your confidence: high, medium, or needs practice.

Chapter: Mixtures, Elements, and Compounds

In this chapter you will study Mixtures, Elements, and Compounds.

Inquiry starter: What do you already know about this topic from home, school, or your community?

Chapter units:

- 2.1 Mixtures
- 2.2 Acids, Bases and Indicators

Before you begin, write two things you already know and one question you want to answer. As you study, collect examples from daily life. At the end of the chapter, return to your question and improve your answer using new vocabulary, examples, and evidence from the lessons.

Mixtures: Lesson Opener

Learning area: Mixtures

Start here: a safe observation at home, school, garden, workshop, health club, or local environment linked to Mixtures.

Curriculum link: This lesson follows the Grade 7 curriculum sequence and focuses on Mixtures.

By the end of this lesson sequence, you should be able to:

- separate homogeneous mixtures using appropriate methods
- outline applications of separating homogeneous mixtures in day-to-day life
- appreciate the use of different methods of separating mixtures in day-to-day life.

Inquiry questions:

- How is the separation of mixtures important in day-to-day life?

Before reading, write one thing you already know and one question you want this lesson to answer.

Mixtures: Learn

Start with this real situation: a safe observation at home, school, garden, workshop, health club, or local environment linked to Mixtures.

Important idea: Mixtures.

Science and technology learning starts with a question. A useful answer is built from evidence: observations, measurements, diagrams, models, tests, data tables, or design trials. Separate what you saw from what you think it means.

A common mistake is writing guesses as facts without observing, testing safely, or recording evidence. Avoid it by separating what you saw from what you think it means.

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

For Grade 7, use labelled sketches, tables, and short evidence-based explanations.

Inline visual support:

| Evidence tool | How to use it in Mixtures |

| --- | --- |

| Labelled sketch | Name visible parts and show direction or change with arrows. |

| Observation table | Record facts before explaining them. |

| Safety note | Name the hazard and the safe action. |

| Conclusion | Begin with: The evidence shows. |

Method for Science and Technology: Observe, ask questions, test safely, record evidence, and explain findings.

Key words:

- Mixtures

Explain the idea in your own words before attempting the activities.

Mixtures: Worked Example

Investigation model for Mixtures: Move from question to evidence.

Question: What evidence would help explain this idea clearly?

Procedure:

1. State the exact question in one sentence.
2. Name the materials, diagram, data, model, or digital tool needed.
3. Decide what will be observed, measured, compared, or designed.
4. Record results in a table, labelled sketch, flow chart, or screenshot note.
5. Explain what the evidence means and name one limit of the investigation.

Evidence table:

| Step | Evidence collected | What it shows |

| --- | --- | --- |

| Observe or test | specific detail | explanation |

Conclusion frame: The evidence supports this answer because _____.

Mixtures: Activity

Activity for Mixtures: Plan a safe observation or investigation.

Inquiry question: How is the separation of mixtures important in day-to-day life?

1. State the question you want to answer.
2. List safe materials and one safety rule.
3. Draw a labelled setup or observation table before starting.
4. Record at least three observations. Do not write guesses as facts.
5. Compare evidence with a partner and discuss what the evidence means.
6. Write a short conclusion that begins, "The evidence shows."

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

Home link: Look for a safe related example at home or in the community. Record what you observed without disturbing people, animals, plants, or devices.

Mixtures: Activity And Practice

Practice for Mixtures:

Use this routine for every answer: question, materials or evidence, method, observation, conclusion, and safety.

1. Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome. Target outcome: separate homogeneous mixtures using appropriate methods. Include labels, evidence or expected observation, a safety note where relevant, and a conclusion.
2. Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome. Target outcome: outline applications of separating homogeneous mixtures in day-to-day life. Include labels, evidence or expected observation, a safety note where relevant, and a conclusion.
3. Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome. Target outcome: appreciate the use of different methods of separating mixtures in day-to-day life. Include labels, evidence or expected observation, a safety note where relevant, and a conclusion.

Correction check:

- observation is specific
- safety is considered
- conclusion uses evidence

Reflection: Which part of your answer is evidence? Which part is explanation?

Home link: Observe a safe related example at home or in the community. Record facts only, without disturbing people, animals, plants, or devices.

Mixtures: Outcome Check

Outcome: separate homogeneous mixtures using appropriate methods

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows." or "The model shows."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Mixtures, uses correct science vocabulary, and connects evidence to explanation.

Mixtures: Outcome Check

Outcome: outline applications of separating homogeneous mixtures in day-to-day life

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows." or "The model shows."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Mixtures, uses correct science vocabulary, and connects evidence to explanation.

Mixtures: Outcome Check

Outcome: appreciate the use of different methods of separating mixtures in day-to-day life.

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows." or "The model shows."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Mixtures, uses correct science vocabulary, and connects evidence to explanation.

Acids, Bases and Indicators: acids and bases using a litmus paper: Lesson Opener

Learning area: Acids, Bases and Indicators: acids and bases using a litmus paper

Start here: a safe observation at home, school, garden, workshop, health club, or local environment linked to Acids, Bases and Indicators: acids and bases using a litmus paper.

Curriculum link: This lesson follows the Grade 7 curriculum sequence and focuses on Acids, Bases and Indicators: acids and bases using a litmus paper.

By the end of this lesson sequence, you should be able to:

- identify acids and bases using a litmus paper
- prepare an acid-base indicator from plant extracts
- describe the physical properties of acids and bases

Inquiry questions:

- Why are acids and bases important?

Before reading, write one thing you already know and one question you want this lesson to answer.

Acids, Bases and Indicators: acids and bases using a litmus paper: Learn

Start with this real situation: a safe observation at home, school, garden, workshop, health club, or local environment linked to Acids, Bases and Indicators: acids and bases using a litmus paper.

Important idea: Acids, Bases and Indicators: acids and bases using a litmus paper.

Science and technology learning starts with a question. A useful answer is built from evidence: observations, measurements, diagrams, models, tests, data tables, or design trials. Separate what you saw from what you think it means.

A common mistake is writing guesses as facts without observing, testing safely, or recording evidence. Avoid it by separating what you saw from what you think it means.

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

For Grade 7, use labelled sketches, tables, and short evidence-based explanations.

Inline visual support:

Evidence tool	How to use it in Acids, Bases and Indicators: acids and bases using a litmus paper
Labelled sketch	Name visible parts and show direction or change with arrows.
Observation table	Record facts before explaining them.
Safety note	Name the hazard and the safe action.
Conclusion	Begin with: The evidence shows.

Method for Science and Technology: Observe, ask questions, test safely, record evidence, and explain findings.

Key words:

- Acids, Bases and Indicators: acids and bases using a litmus paper
- Acids, Bases and Indicators
- Acids
- Bases
- Indicators
- using
- litmus
- paper

Explain the idea in your own words before attempting the activities.

Acids, Bases and Indicators: acids and bases using a litmus paper: Worked Example

Investigation model for Acids, Bases and Indicators: acids and bases using a litmus paper: Move from question to evidence.

Question: What evidence would help explain this idea clearly?

Procedure:

1. State the exact question in one sentence.
2. Name the materials, diagram, data, model, or digital tool needed.
3. Decide what will be observed, measured, compared, or designed.
4. Record results in a table, labelled sketch, flow chart, or screenshot note.
5. Explain what the evidence means and name one limit of the investigation.

Evidence table:

Step	Evidence collected	What it shows
Observe or test	specific detail	explanation

Conclusion frame: The evidence supports this answer because _____.

Acids, Bases and Indicators: acids and bases using a litmus paper: Activity

Activity for Acids, Bases and Indicators: acids and bases using a litmus paper: Plan a safe observation or investigation.

Inquiry question: Why are acids and bases important?

1. State the question you want to answer.
2. List safe materials and one safety rule.
3. Draw a labelled setup or observation table before starting.
4. Record at least three observations. Do not write guesses as facts.
5. Compare evidence with a partner and discuss what the evidence means.
6. Write a short conclusion that begins, "The evidence shows."

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

Home link: Look for a safe related example at home or in the community. Record what you observed without disturbing people, animals, plants, or devices.

Acids, Bases and Indicators: acids and bases using a litmus paper: Activity And Practice

Practice for Acids, Bases and Indicators: acids and bases using a litmus paper:

Use this routine for every answer: question, materials or evidence, method, observation, conclusion, and safety.

1. Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome. Target outcome: identify acids and bases using a litmus paper. Include labels, evidence or expected observation, a safety note where relevant, and a conclusion.
2. Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome. Target outcome: prepare an acid-base indicator from plant extracts. Include labels, evidence or expected observation, a safety note where relevant, and a conclusion.
3. Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome. Target outcome: describe the physical properties of acids and bases. Include labels, evidence or expected observation, a safety note where relevant, and a conclusion.

Correction check:

- observation is specific
- safety is considered
- conclusion uses evidence

Reflection: Which part of your answer is evidence? Which part is explanation?

Home link: Observe a safe related example at home or in the community. Record facts only, without disturbing people, animals, plants, or devices.

Acids, Bases and Indicators: acids and bases using a litmus paper: Outcome Check

Outcome: identify acids and bases using a litmus paper

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows." or "The model shows."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Acids, Bases and Indicators: acids and bases using a litmus paper, uses correct science vocabulary, and connects evidence to explanation.

Acids, Bases and Indicators: acids and bases using a litmus paper: Outcome Check

Outcome: prepare an acid-base indicator from plant extracts

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows." or "The model shows."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Acids, Bases and Indicators: acids and bases using a litmus paper, uses correct science vocabulary, and connects evidence to explanation.

Acids, Bases and Indicators: acids and bases using a litmus paper: Outcome Check

Outcome: describe the physical properties of acids and bases

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows." or "The model shows."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Acids, Bases and Indicators: acids and bases using a litmus paper, uses correct science vocabulary, and connects evidence to explanation.

Acids, Bases and Indicators: outline applications of acids, bases, and indicators: Lesson Opener

Learning area: Acids, Bases and Indicators: outline applications of acids, bases, and indicators

Start here: a safe observation at home, school, garden, workshop, health club, or local environment linked to Acids, Bases and Indicators: outline applications of acids, bases, and indicators.

Curriculum link: This lesson follows the Grade 7 curriculum sequence and focuses on Acids, Bases and Indicators: outline applications of acids, bases, and indicators.

By the end of this lesson sequence, you should be able to:

- outline applications of acids, bases, and indicators in real life
- appreciate the uses of acids and bases in real life.

Inquiry questions:

- Why are acids and bases important?

Before reading, write one thing you already know and one question you want this lesson to answer.

Acids, Bases and Indicators: outline applications of acids, bases, and indicators: Learn

Start with this real situation: a safe observation at home, school, garden, workshop, health club, or local environment linked to Acids, Bases and Indicators: outline applications of acids, bases, and indicators.

Important idea: Acids, Bases and Indicators: outline applications of acids, bases, and indicators.

Science and technology learning starts with a question. A useful answer is built from evidence: observations, measurements, diagrams, models, tests, data tables, or design trials. Separate what you saw from what you think it means.

A common mistake is writing guesses as facts without observing, testing safely, or recording evidence. Avoid it by separating what you saw from what you think it means.

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

For Grade 7, use labelled sketches, tables, and short evidence-based explanations.

Inline visual support:

| Evidence tool | How to use it in Acids, Bases and Indicators: outline applications of acids, bases, and indicators |

| --- | --- |

| Labelled sketch | Name visible parts and show direction or change with arrows. |

| Observation table | Record facts before explaining them. |

| Safety note | Name the hazard and the safe action. |

| Conclusion | Begin with: The evidence shows. |

Method for Science and Technology: Observe, ask questions, test safely, record evidence, and explain findings.

Key words:

- Acids, Bases and Indicators: outline applications of acids, bases, and indicators
- Acids, Bases and Indicators
- Acids
- Bases
- Indicators
- outline
- applications

Explain the idea in your own words before attempting the activities.

Acids, Bases and Indicators: outline applications of acids, bases, and indicators: Worked Example

Investigation model for Acids, Bases and Indicators: outline applications of acids, bases, and indicators: Move from question to evidence.

Question: What evidence would help explain this idea clearly?

Procedure:

1. State the exact question in one sentence.
2. Name the materials, diagram, data, model, or digital tool needed.
3. Decide what will be observed, measured, compared, or designed.
4. Record results in a table, labelled sketch, flow chart, or screenshot note.
5. Explain what the evidence means and name one limit of the investigation.

Evidence table:

Step	Evidence collected	What it shows
---	---	---
Observe or test	specific detail	explanation

Conclusion frame: The evidence supports this answer because _____.

Acids, Bases and Indicators: outline applications of acids, bases, and indicators: Activity

Activity for Acids, Bases and Indicators: outline applications of acids, bases, and indicators: Plan a safe observation or investigation.

Inquiry question: Why are acids and bases important?

1. State the question you want to answer.
2. List safe materials and one safety rule.
3. Draw a labelled setup or observation table before starting.
4. Record at least three observations. Do not write guesses as facts.
5. Compare evidence with a partner and discuss what the evidence means.
6. Write a short conclusion that begins, "The evidence shows."

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

Home link: Look for a safe related example at home or in the community. Record what you observed without disturbing people, animals, plants, or devices.

Acids, Bases and Indicators: outline applications of acids, bases, and indicators: Activity And Practice

Practice for Acids, Bases and Indicators: outline applications of acids, bases, and indicators:

Use this routine for every answer: question, materials or evidence, method, observation, conclusion, and safety.

1. Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome. Target outcome: outline applications of acids, bases, and indicators in real life. Include labels, evidence or expected observation, a safety note where relevant, and a conclusion.

2. Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome. Target outcome: appreciate the uses of acids and bases in real life. Include labels, evidence or expected observation, a safety note where relevant, and a conclusion.

Correction check:

- observation is specific
- safety is considered
- conclusion uses evidence

Reflection: Which part of your answer is evidence? Which part is explanation?

Home link: Observe a safe related example at home or in the community. Record facts only, without disturbing people, animals, plants, or devices.

Acids, Bases and Indicators: outline applications of acids, bases, and indicators: Outcome Check

Outcome: outline applications of acids, bases, and indicators in real life

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows." or "The model shows."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Acids, Bases and Indicators: outline applications of acids, bases, and indicators, uses correct science vocabulary, and connects evidence to explanation.

Acids, Bases and Indicators: outline applications of acids, bases, and indicators: Outcome Check

Outcome: appreciate the uses of acids and bases in real life.

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows." or "The model shows."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Acids, Bases and Indicators: outline applications of acids, bases, and indicators, uses correct science vocabulary, and connects evidence to explanation.

Mixtures, Elements, and Compounds: Chapter Review

Review the chapter carefully.

1. Write five key words from this chapter and explain each one.
2. Choose two units and compare what you learned.
3. Create one poster, chart, dialogue, model, map, or worked solution that teaches this chapter.
4. Answer one inquiry question again. Improve your first answer using what you now know.
5. Rate your confidence: high, medium, or needs practice.

Chapter: Living Things and their Environment

In this chapter you will study Living Things and their Environment.

Inquiry starter: What do you already know about this topic from home, school, or your community?

Chapter units:

- 3.1 Human Reproductive System
- 3.2 Human Excretory System

Before you begin, write two things you already know and one question you want to answer. As you study, collect examples from daily life. At the end of the chapter, return to your question and improve your answer using new vocabulary, examples, and evidence from the lessons.

Human Reproductive System: parts of the male and female reproductive systems: Lesson Opener

Learning area: Human Reproductive System: parts of the male and female reproductive systems

Start here: a safe observation at home, school, garden, workshop, health club, or local environment linked to Human Reproductive System: parts of the male and female reproductive systems.

Curriculum link: This lesson follows the Grade 7 curriculum sequence and focuses on Human Reproductive System: parts of the male and female reproductive systems.

By the end of this lesson sequence, you should be able to:

- identify parts of the male and female reproductive systems
- describe functions of parts of the male and female reproductive system
- describe the physical changes that take place in boys and girls during adolescence

Inquiry questions:

- What are the physical, social, and emotional changes that take place during adolescence?
- How are developmental changes managed during adolescence?

Before reading, write one thing you already know and one question you want this lesson to answer.

Human Reproductive System: parts of the male and female reproductive systems: Learn

Start with this real situation: a safe observation at home, school, garden, workshop, health club, or local environment linked to Human Reproductive System: parts of the male and female reproductive systems.

Important idea: Human Reproductive System: parts of the male and female reproductive systems.

Science and technology learning starts with a question. A useful answer is built from evidence: observations, measurements, diagrams, models, tests, data tables, or design trials. Separate what you saw from what you think it means.

A common mistake is writing guesses as facts without observing, testing safely, or recording evidence. Avoid it by separating what you saw from what you think it means.

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

For Grade 7, use labelled sketches, tables, and short evidence-based explanations.

Inline visual support:

| Evidence tool | How to use it in Human Reproductive System: parts of the male and female reproductive systems |

| --- | --- |

| Labelled sketch | Name visible parts and show direction or change with arrows. |

| Observation table | Record facts before explaining them. |

| Safety note | Name the hazard and the safe action. |

| Conclusion | Begin with: The evidence shows. |

Method for Science and Technology: Observe, ask questions, test safely, record evidence, and explain findings.

Key words:

- Human Reproductive System: parts of the male and female reproductive systems
- Human Reproductive System
- Human
- Reproductive
- System
- parts
- female
- systems

Explain the idea in your own words before attempting the activities.

Human Reproductive System: parts of the male and female reproductive systems: Worked Example

Investigation model for Human Reproductive System: parts of the male and female reproductive systems: Move from question to evidence.

Question: What evidence would help explain this idea clearly?

Procedure:

1. State the exact question in one sentence.
2. Name the materials, diagram, data, model, or digital tool needed.
3. Decide what will be observed, measured, compared, or designed.
4. Record results in a table, labelled sketch, flow chart, or screenshot note.
5. Explain what the evidence means and name one limit of the investigation.

Evidence table:

Step	Evidence collected	What it shows
---	---	---
Observe or test	specific detail	explanation

Conclusion frame: The evidence supports this answer because _____.

Human Reproductive System: parts of the male and female reproductive systems: Activity

Activity for Human Reproductive System: parts of the male and female reproductive systems: Plan a safe observation or investigation.

Inquiry question: What are the physical, social, and emotional changes that take place during adolescence?

1. State the question you want to answer.
2. List safe materials and one safety rule.
3. Draw a labelled setup or observation table before starting.
4. Record at least three observations. Do not write guesses as facts.
5. Compare evidence with a partner and discuss what the evidence means.
6. Write a short conclusion that begins, "The evidence shows."

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

Home link: Look for a safe related example at home or in the community. Record what you observed without disturbing people, animals, plants, or devices.

Human Reproductive System: parts of the male and female reproductive systems: Activity And Practice

Practice for Human Reproductive System: parts of the male and female reproductive systems:

Use this routine for every answer: question, materials or evidence, method, observation, conclusion, and safety.

1. Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome. Target outcome: identify parts of the male and female reproductive systems. Include labels, evidence or expected observation, a safety note where relevant, and a conclusion.
2. Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome. Target outcome: describe functions of parts of the male and female reproductive system. Include labels, evidence or expected observation, a safety note where relevant, and a conclusion.
3. Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome. Target outcome: describe the physical changes that take place in boys and girls during adolescence. Include labels, evidence or expected observation, a safety note where relevant, and a conclusion.

Correction check:

- observation is specific
- safety is considered
- conclusion uses evidence

Reflection: Which part of your answer is evidence? Which part is explanation?

Home link: Observe a safe related example at home or in the community. Record facts only, without disturbing people, animals, plants, or devices.

Human Reproductive System: parts of the male and female reproductive systems: Outcome Check

Outcome: identify parts of the male and female reproductive systems

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows." or "The model shows."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Human Reproductive System: parts of the male and female reproductive systems, uses correct science vocabulary, and connects evidence to explanation.

Human Reproductive System: parts of the male and female reproductive systems: Outcome Check

Outcome: describe functions of parts of the male and female reproductive system

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows." or "The model shows."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Human Reproductive System: parts of the male and female reproductive systems, uses correct science vocabulary, and connects evidence to explanation.

Human Reproductive System: parts of the male and female reproductive systems: Outcome Check

Outcome: describe the physical changes that take place in boys and girls during adolescence

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows." or "The model shows."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Human Reproductive System: parts of the male and female reproductive systems, uses correct science vocabulary, and connects evidence to explanation.

Human Reproductive System: develop a plan to manage developmental challenges during: Lesson Opener

Learning area: Human Reproductive System: develop a plan to manage developmental challenges during

Start here: a safe observation at home, school, garden, workshop, health club, or local environment linked to Human Reproductive System: develop a plan to manage developmental challenges during.

Curriculum link: This lesson follows the Grade 7 curriculum sequence and focuses on Human Reproductive System: develop a plan to manage developmental challenges during.

By the end of this lesson sequence, you should be able to:

- develop a plan to manage developmental challenges during

Inquiry questions:

- How are developmental changes managed during adolescence?

Before reading, write one thing you already know and one question you want this lesson to answer.

Human Reproductive System: develop a plan to manage developmental challenges during: Learn

Start with this real situation: a safe observation at home, school, garden, workshop, health club, or local environment linked to Human Reproductive System: develop a plan to manage developmental challenges during.

Important idea: Human Reproductive System: develop a plan to manage developmental challenges during.

Science and technology learning starts with a question. A useful answer is built from evidence: observations, measurements, diagrams, models, tests, data tables, or design trials. Separate what you saw from what you think it means.

A common mistake is writing guesses as facts without observing, testing safely, or recording evidence. Avoid it by separating what you saw from what you think it means.

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

For Grade 7, use labelled sketches, tables, and short evidence-based explanations.

Inline visual support:

| Evidence tool | How to use it in Human Reproductive System: develop a plan to manage developmental challenges during |

| --- | --- |

| Labelled sketch | Name visible parts and show direction or change with arrows. |

| Observation table | Record facts before explaining them. |

| Safety note | Name the hazard and the safe action. |

| Conclusion | Begin with: The evidence shows. |

Method for Science and Technology: Observe, ask questions, test safely, record evidence, and explain findings.

Key words:

- Human Reproductive System: develop a plan to manage developmental challenges during
- Human Reproductive System
- Human
- Reproductive
- System
- develop
- manage
- developmental

Explain the idea in your own words before attempting the activities.

Human Reproductive System: develop a plan to manage developmental challenges during: Worked Example

Investigation model for Human Reproductive System: develop a plan to manage developmental challenges during: Move from question to evidence.

Question: What evidence would help explain this idea clearly?

Procedure:

1. State the exact question in one sentence.
2. Name the materials, diagram, data, model, or digital tool needed.
3. Decide what will be observed, measured, compared, or designed.
4. Record results in a table, labelled sketch, flow chart, or screenshot note.
5. Explain what the evidence means and name one limit of the investigation.

Evidence table:

Step	Evidence collected	What it shows
---	---	---
Observe or test	specific detail	explanation

Conclusion frame: The evidence supports this answer because _____.

Human Reproductive System: develop a plan to manage developmental challenges during: Activity

Activity for Human Reproductive System: develop a plan to manage developmental challenges during: Plan a safe observation or investigation.

Inquiry question: How are developmental changes managed during adolescence?

1. State the question you want to answer.
2. List safe materials and one safety rule.
3. Draw a labelled setup or observation table before starting.
4. Record at least three observations. Do not write guesses as facts.
5. Compare evidence with a partner and discuss what the evidence means.
6. Write a short conclusion that begins, "The evidence shows."

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

Home link: Look for a safe related example at home or in the community. Record what you observed without disturbing people, animals, plants, or devices.

Human Reproductive System: develop a plan to manage developmental challenges during: Activity And Practice

Practice for Human Reproductive System: develop a plan to manage developmental challenges during:

Use this routine for every answer: question, materials or evidence, method, observation, conclusion, and safety.

1. Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome. Target outcome: develop a plan to manage developmental challenges during. Include labels, evidence or expected observation, a safety note where relevant, and a conclusion.

Correction check:

- observation is specific
- safety is considered
- conclusion uses evidence

Reflection: Which part of your answer is evidence? Which part is explanation?

Home link: Observe a safe related example at home or in the community. Record facts only, without disturbing people, animals, plants, or devices.

Human Reproductive System: develop a plan to manage developmental challenges during: Outcome Check

Outcome: develop a plan to manage developmental challenges during

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows." or "The model shows."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Human Reproductive System: develop a plan to manage developmental challenges during, uses correct science vocabulary, and connects evidence to explanation.

Human Excretory System: functions of parts of the human skin: Lesson Opener

Learning area: Human Excretory System: functions of parts of the human skin

Start here: a safe observation at home, school, garden, workshop, health club, or local environment linked to Human Excretory System: functions of parts of the human skin.

Curriculum link: This lesson follows the Grade 7 curriculum sequence and focuses on Human Excretory System: functions of parts of the human skin.

By the end of this lesson sequence, you should be able to:

- identify functions of parts of the human skin
- identify functions of parts of the urinary system in human
- describe the causes of kidney disorders in human beings

Inquiry questions:

- What is the role of the excretory system?
- How can a healthy excretory system be maintained?

Before reading, write one thing you already know and one question you want this lesson to answer.

Human Excretory System: functions of parts of the human skin: Learn

Start with this real situation: a safe observation at home, school, garden, workshop, health club, or local environment linked to Human Excretory System: functions of parts of the human skin.

Important idea: Human Excretory System: functions of parts of the human skin.

Science and technology learning starts with a question. A useful answer is built from evidence: observations, measurements, diagrams, models, tests, data tables, or design trials. Separate what you saw from what you think it means.

A common mistake is writing guesses as facts without observing, testing safely, or recording evidence. Avoid it by separating what you saw from what you think it means.

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

For Grade 7, use labelled sketches, tables, and short evidence-based explanations.

Inline visual support:

Evidence tool	How to use it in Human Excretory System: functions of parts of the human skin
Labelled sketch	Name visible parts and show direction or change with arrows.
Observation table	Record facts before explaining them.
Safety note	Name the hazard and the safe action.
Conclusion	Begin with: The evidence shows.

Method for Science and Technology: Observe, ask questions, test safely, record evidence, and explain findings.

Key words:

- Human Excretory System: functions of parts of the human skin
- Human Excretory System
- Human
- Excretory
- System
- functions
- parts

Explain the idea in your own words before attempting the activities.

Human Excretory System: functions of parts of the human skin: Worked Example

Investigation model for Human Excretory System: functions of parts of the human skin: Move from question to evidence.

Question: What evidence would help explain this idea clearly?

Procedure:

1. State the exact question in one sentence.
2. Name the materials, diagram, data, model, or digital tool needed.
3. Decide what will be observed, measured, compared, or designed.
4. Record results in a table, labelled sketch, flow chart, or screenshot note.
5. Explain what the evidence means and name one limit of the investigation.

Evidence table:

Step	Evidence collected	What it shows
---	---	---
Observe or test	specific detail	explanation

Conclusion frame: The evidence supports this answer because _____.

Human Excretory System: functions of parts of the human skin: Activity

Activity for Human Excretory System: functions of parts of the human skin: Plan a safe observation or investigation.

Inquiry question: What is the role of the excretory system?

1. State the question you want to answer.
2. List safe materials and one safety rule.
3. Draw a labelled setup or observation table before starting.
4. Record at least three observations. Do not write guesses as facts.
5. Compare evidence with a partner and discuss what the evidence means.
6. Write a short conclusion that begins, "The evidence shows."

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

Home link: Look for a safe related example at home or in the community. Record what you observed without disturbing people, animals, plants, or devices.

Human Excretory System: functions of parts of the human skin: Activity And Practice

Practice for Human Excretory System: functions of parts of the human skin:

Use this routine for every answer: question, materials or evidence, method, observation, conclusion, and safety.

1. Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome. Target outcome: identify functions of parts of the human skin. Include labels, evidence or expected observation, a safety note where relevant, and a conclusion.
2. Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome. Target outcome: identify functions of parts of the urinary system in human. Include labels, evidence or expected observation, a safety note where relevant, and a conclusion.
3. Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome. Target outcome: describe the causes of kidney disorders in human beings. Include labels, evidence or expected observation, a safety note where relevant, and a conclusion.

Correction check:

- observation is specific
- safety is considered
- conclusion uses evidence

Reflection: Which part of your answer is evidence? Which part is explanation?

Home link: Observe a safe related example at home or in the community. Record facts only, without disturbing people, animals, plants, or devices.

Human Excretory System: functions of parts of the human skin: Outcome Check

Outcome: identify functions of parts of the human skin

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows." or "The model shows."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Human Excretory System: functions of parts of the human skin, uses correct science vocabulary, and connects evidence to explanation.

Human Excretory System: functions of parts of the human skin: Outcome Check

Outcome: identify functions of parts of the urinary system in human

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows." or "The model shows."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Human Excretory System: functions of parts of the human skin, uses correct science vocabulary, and connects evidence to explanation.

Human Excretory System: functions of parts of the human skin: Outcome Check

Outcome: describe the causes of kidney disorders in human beings

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows." or "The model shows."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Human Excretory System: functions of parts of the human skin, uses correct science vocabulary, and connects evidence to explanation.

Human Excretory System: develop and maintain a daily log of activities: Lesson Opener

Learning area: Human Excretory System: develop and maintain a daily log of activities

Start here: a safe observation at home, school, garden, workshop, health club, or local environment linked to Human Excretory System: develop and maintain a daily log of activities.

Curriculum link: This lesson follows the Grade 7 curriculum sequence and focuses on Human Excretory System: develop and maintain a daily log of activities.

By the end of this lesson sequence, you should be able to:

- develop and maintain a daily log of activities that promote skin and kidney health in human beings
- appreciate the need for a healthy lifestyle to promote kidney and skin health.

Inquiry questions:

- How can a healthy excretory system be maintained?

Before reading, write one thing you already know and one question you want this lesson to answer.

Human Excretory System: develop and maintain a daily log of activities: Learn

Start with this real situation: a safe observation at home, school, garden, workshop, health club, or local environment linked to Human Excretory System: develop and maintain a daily log of activities.

Important idea: Human Excretory System: develop and maintain a daily log of activities.

Science and technology learning starts with a question. A useful answer is built from evidence: observations, measurements, diagrams, models, tests, data tables, or design trials. Separate what you saw from what you think it means.

A common mistake is writing guesses as facts without observing, testing safely, or recording evidence. Avoid it by separating what you saw from what you think it means.

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

For Grade 7, use labelled sketches, tables, and short evidence-based explanations.

Inline visual support:

| Evidence tool | How to use it in Human Excretory System: develop and maintain a daily log of activities |

| --- | --- |

| Labelled sketch | Name visible parts and show direction or change with arrows. |

| Observation table | Record facts before explaining them. |

| Safety note | Name the hazard and the safe action. |

| Conclusion | Begin with: The evidence shows. |

Method for Science and Technology: Observe, ask questions, test safely, record evidence, and explain findings.

Key words:

- Human Excretory System: develop and maintain a daily log of activities
- Human Excretory System
- Human
- Excretory
- System
- develop
- maintain

Explain the idea in your own words before attempting the activities.

Human Excretory System: develop and maintain a daily log of activities: Worked Example

Investigation model for Human Excretory System: develop and maintain a daily log of activities: Move from question to evidence.

Question: What evidence would help explain this idea clearly?

Procedure:

1. State the exact question in one sentence.
2. Name the materials, diagram, data, model, or digital tool needed.
3. Decide what will be observed, measured, compared, or designed.
4. Record results in a table, labelled sketch, flow chart, or screenshot note.
5. Explain what the evidence means and name one limit of the investigation.

Evidence table:

Step	Evidence collected	What it shows
---	---	---
Observe or test	specific detail	explanation

Conclusion frame: The evidence supports this answer because _____.

Human Excretory System: develop and maintain a daily log of activities: Activity

Activity for Human Excretory System: develop and maintain a daily log of activities: Plan a safe observation or investigation.

Inquiry question: How can a healthy excretory system be maintained?

1. State the question you want to answer.
2. List safe materials and one safety rule.
3. Draw a labelled setup or observation table before starting.
4. Record at least three observations. Do not write guesses as facts.
5. Compare evidence with a partner and discuss what the evidence means.
6. Write a short conclusion that begins, "The evidence shows."

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

Home link: Look for a safe related example at home or in the community. Record what you observed without disturbing people, animals, plants, or devices.

Human Excretory System: develop and maintain a daily log of activities: Activity And Practice

Practice for Human Excretory System: develop and maintain a daily log of activities:

Use this routine for every answer: question, materials or evidence, method, observation, conclusion, and safety.

1. Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome. Target outcome: develop and maintain a daily log of activities that promote skin and kidney health in human beings. Include labels, evidence or expected observation, a safety note where relevant, and a conclusion.
2. Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome. Target outcome: appreciate the need for a healthy lifestyle to promote kidney and skin health. Include labels, evidence or expected observation, a safety note where relevant, and a conclusion.

Correction check:

- observation is specific
- safety is considered
- conclusion uses evidence

Reflection: Which part of your answer is evidence? Which part is explanation?

Home link: Observe a safe related example at home or in the community. Record facts only, without disturbing people, animals, plants, or devices.

Human Excretory System: develop and maintain a daily log of activities: Outcome Check

Outcome: develop and maintain a daily log of activities that promote skin and kidney health in human beings

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows." or "The model shows."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Human Excretory System: develop and maintain a daily log of activities, uses correct science vocabulary, and connects evidence to explanation.

Human Excretory System: develop and maintain a daily log of activities: Outcome Check

Outcome: appreciate the need for a healthy lifestyle to promote kidney and skin health.

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows." or "The model shows."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Human Excretory System: develop and maintain a daily log of activities, uses correct science vocabulary, and connects evidence to explanation.

Living Things and their Environment: Chapter Review

Review the chapter carefully.

1. Write five key words from this chapter and explain each one.
2. Choose two units and compare what you learned.
3. Create one poster, chart, dialogue, model, map, or worked solution that teaches this chapter.
4. Answer one inquiry question again. Improve your first answer using what you now know.
5. Rate your confidence: high, medium, or needs practice.

Chapter: Force and Energy

In this chapter you will study Force and Energy.

Inquiry starter: What do you already know about this topic from home, school, or your community?

Chapter units:

- 4.1 Electrical
- 4.2 Magnetism

Before you begin, write two things you already know and one question you want to answer. As you study, collect examples from daily life. At the end of the chapter, return to your question and improve your answer using new vocabulary, examples, and evidence from the lessons.

Electrical: Lesson Opener

Learning area: Electrical

Start here: a safe observation at home, school, garden, workshop, health club, or local environment linked to Electrical.

Curriculum link: This lesson follows the Grade 7 curriculum sequence and focuses on Electrical.

By the end of this lesson sequence, you should be able to:

- electricity in the environment, electrical circuits to demonstrate the flow of electric current, electrical appliances used

Inquiry questions:

- What are the sources of electricity in the environment?
- How is electricity used in day-to-day life?

Before reading, write one thing you already know and one question you want this lesson to answer.

Electrical: Learn

Start with this real situation: a safe observation at home, school, garden, workshop, health club, or local environment linked to Electrical.

Important idea: Electrical.

Physical science looks for patterns in energy, forces, and materials. Keep one condition changing at a time, observe the effect, and use measured evidence where possible. A correct explanation names the cause, the change observed, and the reason the change happened.

A common mistake is writing guesses as facts without observing, testing safely, or recording evidence. Avoid it by separating what you saw from what you think it means.

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

For Grade 7, use labelled sketches, tables, and short evidence-based explanations.

Inline visual support:

| Evidence tool | How to use it in Electrical |

| --- | --- |

| Labelled sketch | Name visible parts and show direction or change with arrows. |

| Observation table | Record facts before explaining them. |

| Safety note | Name the hazard and the safe action. |

| Conclusion | Begin with: The evidence shows. |

Method for Science and Technology: Observe, ask questions, test safely, record evidence, and explain findings.

Key words:

- Electrical

Explain the idea in your own words before attempting the activities.

Electrical: Worked Example

Investigation model for Electrical: Test one energy or circuit condition safely.

Question: What changes when one condition is changed?

Example: In a simple circuit, a bulb lights only when there is a complete path from one battery terminal, through the bulb, and back to the other terminal.

Procedure:

1. Name the variable you will change, such as number of cells, wire length, material, distance, or surface.
2. Name what you will observe or measure.
3. Keep the other conditions the same.
4. Record results in a table with at least three trials or comparisons.
5. Write a conclusion that mentions the pattern in the results.

Safety rule: Use only approved school materials and ask before connecting any electrical device.

Electrical: Activity

Activity for Electrical: Plan a safe observation or investigation.

Inquiry question: What are the sources of electricity in the environment?

1. State the question you want to answer.
2. List safe materials and one safety rule.
3. Draw a labelled setup or observation table before starting.
4. Record at least three observations. Do not write guesses as facts.
5. Compare evidence with a partner and discuss what the evidence means.
6. Write a short conclusion that begins, "The evidence shows."

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

Home link: Look for a safe related example at home or in the community. Record what you observed without disturbing people, animals, plants, or devices.

Electrical: Activity And Practice

Practice for Electrical:

Use this routine for every answer: question, materials or evidence, method, observation, conclusion, and safety.

1. Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome. Target outcome: electricity in the environment, electrical circuits to demonstrate the flow of electric current, electrical appliances used. Include labels, evidence or expected observation, a safety note where relevant, and a conclusion.

Correction check:

- observation is specific
- safety is considered
- conclusion uses evidence

Reflection: Which part of your answer is evidence? Which part is explanation?

Home link: Observe a safe related example at home or in the community. Record facts only, without disturbing people, animals, plants, or devices.

Electrical: Outcome Check

Outcome: electricity in the environment, electrical circuits to demonstrate the flow of electric current, electrical appliances used

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows." or "The model shows."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Electrical, uses correct science vocabulary, and connects evidence to explanation.

Magnetism: Lesson Opener

Learning area: Magnetism

Start here: a safe observation at home, school, garden, workshop, health club, or local environment linked to Magnetism.

Curriculum link: This lesson follows the Grade 7 curriculum sequence and focuses on Magnetism.

By the end of this lesson sequence, you should be able to:

- properties of a magnet, magnetic or non-magnetic, magnets in day-to-day life, applications of magnets in day-to-day life

Inquiry questions:

- How are magnets used in day-to-day life?

Before reading, write one thing you already know and one question you want this lesson to answer.

Magnetism: Learn

Start with this real situation: a safe observation at home, school, garden, workshop, health club, or local environment linked to Magnetism.

Important idea: Magnetism.

Science and technology learning starts with a question. A useful answer is built from evidence: observations, measurements, diagrams, models, tests, data tables, or design trials. Separate what you saw from what you think it means.

A common mistake is writing guesses as facts without observing, testing safely, or recording evidence. Avoid it by separating what you saw from what you think it means.

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

For Grade 7, use labelled sketches, tables, and short evidence-based explanations.

Inline visual support:

| Evidence tool | How to use it in Magnetism |

| --- | --- |

| Labelled sketch | Name visible parts and show direction or change with arrows. |

| Observation table | Record facts before explaining them. |

| Safety note | Name the hazard and the safe action. |

| Conclusion | Begin with: The evidence shows. |

Method for Science and Technology: Observe, ask questions, test safely, record evidence, and explain findings.

Key words:

- Magnetism

Explain the idea in your own words before attempting the activities.

Magnetism: Worked Example

Investigation model for Magnetism: Map a magnetic field safely.

Question: Where is a magnet's field strongest?

Example: Iron filings gather most closely near the north and south poles of a bar magnet. A compass needle changes direction when moved around the magnet because it aligns with the magnetic field.

Procedure:

1. Place a bar magnet under a sheet of paper.
2. Sprinkle a small amount of iron filings or use a compass at different positions.
3. Sketch the field pattern and label north pole, south pole, and field direction.
4. Compare the spacing of field lines near and far from the poles.
5. For induction, explain that a changing magnetic field can produce an electric current in a conductor.

Conclusion frame: The magnetic field is strongest where _____.

Magnetism: Activity

Activity for Magnetism: Plan a safe observation or investigation.

Inquiry question: How are magnets used in day-to-day life?

1. State the question you want to answer.
2. List safe materials and one safety rule.
3. Draw a labelled setup or observation table before starting.
4. Record at least three observations. Do not write guesses as facts.
5. Compare evidence with a partner and discuss what the evidence means.
6. Write a short conclusion that begins, "The evidence shows."

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

Home link: Look for a safe related example at home or in the community. Record what you observed without disturbing people, animals, plants, or devices.

Magnetism: Activity And Practice

Practice for Magnetism:

Use this routine for every answer: question, materials or evidence, method, observation, conclusion, and safety.

1. Draw a bar magnet with field lines from north to south, show where the field is strongest, and describe one test using iron filings or a compass. Answer guide: field lines are closest near the poles, a compass aligns with the field, and electromagnetic induction needs a changing magnetic field. Target outcome: properties of a magnet, magnetic or non-magnetic, magnets in day-to-day life, applications of magnets in day-to-day life. Include labels, evidence or expected observation, a safety note where relevant, and a conclusion.

Correction check:

- observation is specific
- safety is considered
- conclusion uses evidence

Reflection: Which part of your answer is evidence? Which part is explanation?

Home link: Observe a safe related example at home or in the community. Record facts only, without disturbing people, animals, plants, or devices.

Magnetism: Outcome Check

Outcome: properties of a magnet, magnetic or non-magnetic, magnets in day-to-day life, applications of magnets in day-to-day life

Show mastery with topic evidence.

Task: Draw a bar magnet with field lines from north to south, show where the field is strongest, and describe one test using iron filings or a compass. Answer guide: field lines are closest near the poles, a compass aligns with the field, and electromagnetic induction needs a changing magnetic field.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows." or "The model shows."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Magnetism, uses correct science vocabulary, and connects evidence to explanation.

Force and Energy: Chapter Review

Review the chapter carefully.

1. Write five key words from this chapter and explain each one.
2. Choose two units and compare what you learned.
3. Create one poster, chart, dialogue, model, map, or worked solution that teaches this chapter.
4. Answer one inquiry question again. Improve your first answer using what you now know.
5. Rate your confidence: high, medium, or needs practice.

Introduction to Integrated Science: Review Clinic 1

Use this review clinic to strengthen a real unit from the book.

Practice context: home or school example connected to Introduction to Integrated Science.

1. Re-read the lesson on Introduction to Integrated Science and underline the main idea.
2. Write a five-sentence or five-step summary.
3. Make one question that tests knowledge.
4. Make one question that tests skill, reasoning, or application.
5. Solve or answer both questions.
6. Check your work against this criterion: observation is specific.
7. Exchange your questions with a partner and discuss corrections.

Challenge: Describe a real-life task from your home, school, farm, market, field, library, or community that uses this idea.

Laboratory Safety: in the laboratory and access: Review Clinic 2

Use this review clinic to strengthen a real unit from the book.

Practice context: county or community example connected to Laboratory Safety: in the laboratory and access.

1. Re-read the lesson on Laboratory Safety: in the laboratory and access and underline the main idea.
2. Write a five-sentence or five-step summary.
3. Make one question that tests knowledge.
4. Make one question that tests skill, reasoning, or application.
5. Solve or answer both questions.
6. Check your work against this criterion: safety is considered.
7. Exchange your questions with a partner and discuss corrections.

Challenge: Describe a real-life task from your home, school, farm, market, field, library, or community that uses this idea.

Laboratory apparatus and instruments: basic practise use and care for apparatus skills: Review Clinic 3

Use this review clinic to strengthen a real unit from the book.

Practice context: partner discussion about Laboratory apparatus and instruments: basic practise use and care for apparatus skills.

1. Re-read the lesson on Laboratory apparatus and instruments: basic practise use and care for apparatus skills and underline the main idea.
2. Write a five-sentence or five-step summary.
3. Make one question that tests knowledge.
4. Make one question that tests skill, reasoning, or application.
5. Solve or answer both questions.
6. Check your work against this criterion: conclusion uses evidence.
7. Exchange your questions with a partner and discuss corrections.

Challenge: Describe a real-life task from your home, school, farm, market, field, library, or community that uses this idea.

Mixtures: Review Clinic 4

Use this review clinic to strengthen a real unit from the book.

Practice context: small project or observation linked to Mixtures.

1. Re-read the lesson on Mixtures and underline the main idea.
2. Write a five-sentence or five-step summary.
3. Make one question that tests knowledge.
4. Make one question that tests skill, reasoning, or application.
5. Solve or answer both questions.
6. Check your work against this criterion: observation is specific.
7. Exchange your questions with a partner and discuss corrections.

Challenge: Describe a real-life task from your home, school, farm, market, field, library, or community that uses this idea.

Glossary

Use this glossary to revise important words in Science and Technology.

- Introduction to Integrated Science: A science idea to explain with observation, labelled diagrams, safe investigation, evidence, and a conclusion.
- Laboratory Safety: A science idea to explain with observation, labelled diagrams, safe investigation, evidence, and a conclusion.
- Laboratory apparatus and instruments: A science idea to explain with observation, labelled diagrams, safe investigation, evidence, and a conclusion.
- Mixtures: A science idea to explain with observation, labelled diagrams, safe investigation, evidence, and a conclusion.
- Acids, Bases and Indicators: A science idea to explain with observation, labelled diagrams, safe investigation, evidence, and a conclusion.
- Human Reproductive System: A science idea to explain with observation, labelled diagrams, safe investigation, evidence, and a conclusion.
- Human Excretory System: A science idea to explain with observation, labelled diagrams, safe investigation, evidence, and a conclusion.
- Electrical: A science idea to explain with observation, labelled diagrams, safe investigation, evidence, and a conclusion.
- Magnetism: A science idea to explain with observation, labelled diagrams, safe investigation, evidence, and a conclusion.

Add five more words from your lessons, then write a meaning and one correct example sentence or worked example for each word.

Answer And Teacher Notes

A strong Science and Technology answer should:

- answer the exact question asked;
- separate observation from explanation;
- name materials, variables, safety checks, or data sources where needed;
- use labelled diagrams, tables, or step-by-step procedures when useful;
- draw a conclusion from evidence, not from guessing.

For investigations, good answers include a question, method, results, conclusion, and one limit or improvement. For technology tasks, good answers explain the design choice and how it solves the problem safely.

Answer-key guide: a complete science answer names the concept, shows evidence in a labelled sketch/table/model, explains the pattern, includes safety where relevant, and ends with a conclusion beginning from the evidence. Wrong answers often mix observation with opinion or leave out variables.

Rubric for self-checking:

- 4: The answer addresses the exact question, uses subject vocabulary correctly, gives evidence or working, and includes a useful check or correction.
- 3: The main answer is correct but one part needs stronger evidence, clearer working, better labels, or a fuller explanation.
- 2: Some understanding is shown, but the answer is incomplete, too general, or hard to follow.
- 1: The answer is guessed, off topic, or missing the evidence needed for the task.

Correction routine: reread the command word, underline the key data or evidence, improve one weak step, and write one sentence explaining why the final answer is reasonable.

Final Project

Create a final Science and Technology project.

Your project must include:

1. A clear question, problem, or design need.
2. Evidence from at least three lessons.
3. A safe method, labelled diagram, data table, flow chart, model, or prototype plan.
4. A conclusion that explains what the evidence shows.
5. One safety note, one limitation, and one improvement.

Present your project to a classmate, teacher, parent, or study group.